BIOL-1 Practice Test 01 — Answer Sheet

Modules 1-6: Study of Life through Metabolism

Name:		Date:		Score:	

✂ **TEAR-OFF ANSWER SHEET:** record every Part A: Multiple Choice (25 questions) and Part B: Fill in the Blank (10 questions) answer on this page. Free response begins on the reverse side. Detach and turn in this sheet.

Part A: Multiple Choice (25 questions)

1.		10.		19.	
2.		11.		20.	
3.		12.		21.	
4.		13.		22.	
5.		14.		23.	
6.		15.		24.	
7.		16.		25.	
8.		17.			
9.		18.			

Part B: Fill in the Blank (10 questions)

26.		31.	
27.		32.	
28.		33.	
29.		34.	
30.		35.	

BIOL-1 Practice Test 01 — Part C: Short Answer (4 questions)

Answer each question in 2-3 complete sentences.

36. Name three characteristics that all living things share and briefly describe each one.

37. What are the four major types of biological macromolecules? Name the monomer (building block) for proteins.

38. Explain the difference between passive transport and active transport. Give one example of each.

39. Explain how enzymes speed up chemical reactions and why they are specific to their substrates.

Response 1 — Question Number: _____

Response 2 — Question Number: _____

Response 3 — Question Number: _____

Response 4 — Question Number: _____

BIOL-1 Practice Test 01 — Question Booklet

Answer Parts A and B on the tear-off sheet. Keep this booklet with your exam materials.

Part A: Multiple Choice (25 questions)

Choose the best answer for each question.

Module 1: Study of Life

1. Which of the following is a characteristic shared by ALL living organisms?
    - A) Uses photosynthesis for energy
    - B) Movement from place to place
    - C) Made of one or more cells
    - D) Reproduces sexually

2. In the scientific method, what comes AFTER making an observation?
    - A) Drawing a conclusion
    - B) Forming a hypothesis
    - C) Conducting an experiment
    - D) Publishing results

3. In a controlled experiment, the variable that the scientist changes is called the:
    - A) Independent variable
    - B) Constant
    - C) Control variable
    - D) Dependent variable

Module 2: Basic Chemistry

4. What determines the atomic number of an element?
    - A) Number of protons
    - B) Number of neutrons
    - C) Atomic mass
    - D) Number of electrons

5. When electrons are TRANSFERRED between atoms, what type of bond is formed?
    - A) Covalent bond
    - B) Ionic bond
    - C) Hydrogen bond
    - D) Polar bond

6. Water is considered a polar molecule because:
    - A) It has a high specific heat
    - B) It can dissolve many substances
    - C) Electrons are shared unequally between atoms
    - D) It contains hydrogen atoms

7. A substance that resists changes in pH is called a:
    - A) Acid
    - B) Buffer
    - C) Base
    - D) Solvent

Module 3: Organic Molecules

8. Which element is the "backbone" of all organic molecules?
    - A) Nitrogen
    - B) Oxygen
    - C) Hydrogen
    - D) Carbon
9. What is the relationship between monomers and polymers?
    - A) Polymers are the building blocks of monomers
    - B) Monomers break down polymers
    - C) Monomers link together to form polymers
    - D) They are the same thing
10. Which biomolecule is the body's primary source of quick energy?
    - A) Protein
    - B) Nucleic acid
    - C) Lipid
    - D) Carbohydrate
11. What type of bond holds amino acids together in a protein?
    - A) Glycosidic bond
    - B) Hydrogen bond
    - C) Ionic bond
    - D) Peptide bond

Module 4: Cells

12. According to cell theory, all cells:
    - A) Contain a nucleus
    - B) Come from pre-existing cells
    - C) Are identical in structure
    - D) Can survive independently
13. The main difference between prokaryotic and eukaryotic cells is that prokaryotes:
    - A) Are larger than eukaryotes
    - B) Have more organelles
    - C) Cannot reproduce
    - D) Lack a membrane-bound nucleus
14. Which organelle is called the "powerhouse of the cell" because it produces ATP?
    - A) Mitochondria
    - B) Golgi apparatus
    - C) Nucleus
    - D) Ribosome
15. Which structure is found in plant cells but NOT in animal cells?
    - A) Mitochondria
    - B) Ribosomes
    - C) Nucleus
    - D) Cell wall

Module 5: Membranes

16. The cell membrane is best described by the:
    - A) Fluid mosaic model
    - B) Rigid barrier model
    - C) Lock and key model
    - D) Simple lipid model

17. The cell membrane is primarily composed of:
    - A) Proteins only
    - B) Phospholipid bilayer
    - C) Carbohydrates
    - D) Nucleic acids

18. Phospholipids spontaneously form a bilayer in water because:
    - A) They are charged molecules
    - B) They have hydrophilic heads and hydrophobic tails
    - C) They are completely hydrophobic
    - D) They are all the same size

19. Which transport process requires ATP?
    - A) Diffusion
    - B) Facilitated diffusion
    - C) Osmosis
    - D) Active transport

20. When a cell is placed in a hypotonic solution, water will:
    - A) Move into the cell
    - B) Not move
    - C) Move in both directions equally
    - D) Move out of the cell

Module 6: Metabolism

21. Enzymes function by:
    - A) Providing energy for reactions
    - B) Decreasing activation energy
    - C) Increasing activation energy
    - D) Changing the equilibrium of reactions

22. The molecule that stores and transfers energy in all cells is:
    - A) Glucose
    - B) DNA
    - C) ATP
    - D) NADH

23. The part of an enzyme where the substrate binds is called the:
    - A) Active site
    - B) Product site
    - C) Inhibitor site
    - D) Allosteric site

24. When an enzyme loses its shape due to heat or pH changes, this is called:
    - A) Activation
    - B) Catalysis
    - C) Inhibition
    - D) Denaturation

25. Which statement about enzymes is TRUE?
    - A) Enzymes are made of carbohydrates
    - B) Enzymes increase the activation energy
    - C) Enzymes are used up in reactions
    - D) Enzymes can be reused

Part B: Fill in the Blank (10 questions)

Write the correct term in the blank.

26. The basic unit of life is the _____.

27. The _____ is the organelle that contains the cell's genetic material (DNA).

28. The process of breaking down polymers by adding water is called _____.

29. A solution with a pH less than 7 is _____.

30. The monomer (building block) of proteins is the _____.

31. Organisms that cannot make their own food and must consume other organisms are called _____.

32. According to _____ theory, mitochondria and chloroplasts were once free-living bacteria.

33. The process by which water moves across a membrane is called _____.

34. The specific reactant that an enzyme acts on is called the _____.

35. The sum of all chemical reactions in an organism is called _____.
